TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Dyeing with Reactive, Sulphur, Azoic and Disperse Dyes and Pigments

Five coloration chemistries on cotton and polyester — combination shades, shade matching, and stripping — carried out and recorded as one integrated sessional.

WP-4102 · TEXTILE CHEMISTRY & COLORATION-III SESSIONAL · EXPERIMENT 1 · SPECIALIZATION — WET PROCESSING

1. Aim and theory

Aim: to dye textile substrates with five major colorant systems — **reactive dyes** and **sulphur dyes** and **azoic colours** on cotton, **disperse dyes** on polyester by the high-temperature (HT) method, and **pigments** with a binder system on cotton — and then to extend the work to **combination shade dyeing**, a systematic **shade matching** workflow, and **reductive stripping** of a faulty dyeing. The student must be able to explain, for each class, why the recipe, temperature profile and aftertreatment are what they are.

Why five classes in one experiment? No single dye class colours every fibre well. Each class rests on a different fixation principle, and the choice of class is therefore dictated first by the fibre and only second by the shade. Reactive dyes form covalent bonds with cellulose hydroxyl groups and give the highest wet fastness on cotton. Sulphur dyes are reduced to a soluble leuco form, exhausted onto cotton and oxidised back inside the fibre, giving cheap, deep, dull shades. Azoic colours are not applied as dyes at all: two soluble components meet inside the fibre and couple to form an insoluble azo pigment there. Disperse dyes are sparingly soluble, non-ionic colorants that dissolve molecularly into hydrophobic polyester at high temperature. Pigments have no affinity for any fibre; they are glued to the surface by a polymer binder film that is then cured. Understanding these five mechanisms covers almost every coloration decision a wet-processing technologist meets in a dyehouse.

CLASS	FIBRE IN THIS SHEET	FIXATION PRINCIPLE	KEY CONDITIONS / CHARACTER
Reactive	Cotton	Covalent bond dye–O–cellulose under alkali	Salt for exhaustion, soda ash for fixation; 60 °C; excellent wet fastness

Sulphur	Cotton	Reduction to soluble leuco, then oxidation in fibre	Na_2S reduction; near boil; cheap blacks/browns, moderate fastness
Azoic	Cotton	Insoluble azo pigment formed in situ by coupling	Naphtholation, then diazotised base; brilliant reds; high wet fastness
Disperse	Polyester	Solid solution in hydrophobic fibre	pH 4.5–5.0, 130 °C HT; reduction clearing after dyeing
Pigment	Cotton (any fibre)	Binder film mechanically holds pigment	Pad/print, dry, cure 150 °C; shade independent of fibre; handle affected

Three practical consequences follow directly. First, **temperature**: cotton dyeings run at 60–98 °C in open vessels, whereas polyester needs 130 °C and therefore a sealed HT beaker or machine. Second, **liquor control**: reactive and sulphur baths are alkaline and reducing respectively, while the disperse bath is mildly acidic — baths must never be interchanged without thorough washing. Third, **aftertreatment**: soaping off hydrolysed reactive dye, oxidising sulphur dyeings, and reduction clearing of polyester are not optional finishing touches; they are the steps that deliver the rated fastness.

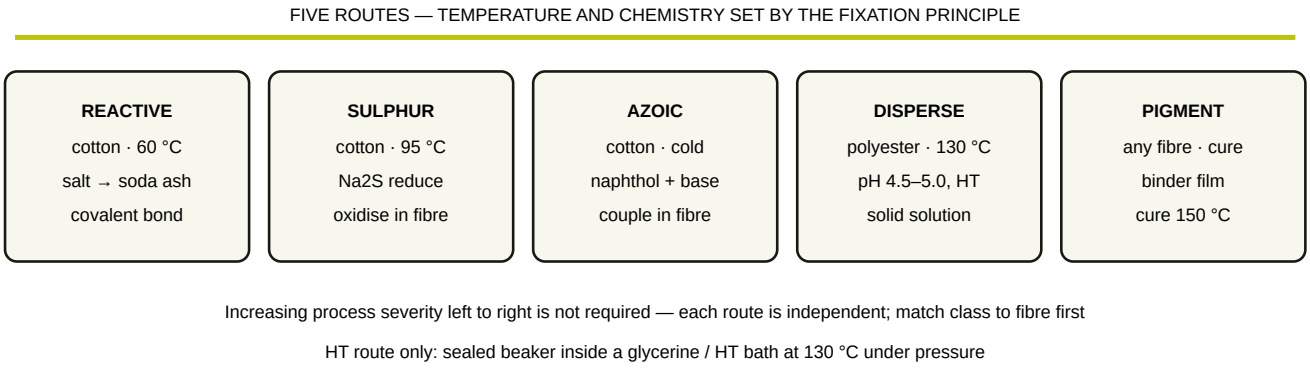


Fig. 1 — The five dyeing routes of this experiment, with their working temperatures and fixation chemistry.

2. Apparatus and materials

Balance (0.01 g), measuring cylinders, pipettes, beakers (500 mL), glass rods, thermometer / temperature controller, laboratory dyeing bath or water bath, sealed stainless-steel HT beakers with HT dyeing machine or glycerine bath for the disperse dyeing, padding mangle or flat glass rod for pigment application, hot-air oven or curing chamber (150 °C), pH paper or pH meter, standard multifibre strip (optional, for staining observation), scoured and bleached cotton fabric, scoured polyester fabric, distilled water. Dyes and chemicals as listed in each procedure. All cotton must be fully scoured and

absorbent (absorbency under 3 s by drop test); polyester must be scoured to remove spin finishes, which otherwise cause unlevel disperse dyeing.

> **LIQUOR RATIO — READ BEFORE STARTING**
>
> LR is stated for every bath in this sheet and must not be changed casually. Raising the LR dilutes salt and alkali (g/L values then need more total chemical) and lowers exhaustion; lowering it risks unlevelness. All worked amounts below are calculated for the stated LR only.

3. Procedure A — reactive dye on cotton

Recipe: reactive dye 2.0% owf; Glauber's salt 40 g/L; soda ash 15 g/L; wetting agent 1 g/L; LR 1:20; 60 °C. The dye used here is a hot-brand (high-reactivity conditions as directed by the supplier; cold-brand dyes run at 40–50 °C with the same sequence).

1. Weigh 10 g scoured cotton. Prepare a 200 mL bath (LR 1:20). Dissolve the dye separately in hot water and add through a strainer.
2. Enter the pre-wetted fabric at 30 °C. Run 10 min in dye and wetting agent only — this is the level-entry stage.
3. Add Glauber's salt 40 g/L in two portions 10 min apart. Salt suppresses the negative surface charge of cellulose in water so the anionic dye can approach and exhaust.
4. Raise to 60 °C at 1.5 °C/min and run 15 min for migration and levelling.
5. Add soda ash 15 g/L in two portions. Alkali ionises cellulose hydroxyl groups (Cell–O^-) which attack the reactive group of the dye, forming the covalent bond; a competing reaction with water produces hydrolysed dye that can never fix. Run 45 min at 60 °C.
6. Drain, rinse cold, neutralise with acetic acid 0.5 mL/L if the fabric is alkaline, then soap at the boil with 2 g/L detergent for 10 min to remove hydrolysed and unfixed dye, rinse and dry. Skipping the boil-soaping is the commonest cause of poor wash fastness in student reactive dyeings.

$$\text{Cell–OH} + \text{HO}^- \rightarrow \text{Cell–O}^- \cdot \text{Cell–O}^- + \text{Dye–X} \rightarrow \text{Cell–O–Dye} + \text{X}^- \cdot \text{H}_2\text{O} + \text{Dye–X} \rightarrow \text{HO–Dye (hydrolysed, unfixed)}$$

4. Procedure B — sulphur dye on cotton

Recipe: sulphur dye 3% owf; sodium sulphide 3% owf (reducing / solubilising agent); soda ash 2 g/L; common salt 10 g/L; LR 1:20; 90–95 °C.

1. Paste 0.30 g sulphur dye with a little soda ash solution and hot water. Add sodium sulphide 0.30 g dissolved in hot water and digest at 60–70 °C for 10 min until the dye dissolves to its leuco form.

Sulphur dyes are held together by disulphide (–S–S–) linkages; sulphide cleaves them to soluble thiol (–SH / –SNa) groups.

2. Make up the bath to 200 mL with soda ash 2 g/L (0.40 g total in bath). Enter pre-wetted 10 g cotton at 50 °C.
3. Raise to 90–95 °C, add salt 10 g/L (2.0 g) in two portions, and run 45 min. Keep the bath covered and the material submerged: air re-oxidises leuco dye at the surface (bronziness).
4. Drain, rinse until the water is nearly clear, then **oxidise**: treat 15 min at 50 °C with hydrogen peroxide 2 mL/L or sodium perborate 2 g/L to reform the insoluble disulphide pigment inside the fibre.
5. Soap mildly (1 g/L, 60 °C, 10 min), rinse and dry away from prolonged strong sunlight for the record sample. Record the characteristically dull, deep shade and note the sulphurous odour of the wet bath.

Note on tendering: sulphur-dyed cotton, especially sulphur black, can tender on storage because residual sulphur slowly oxidises to sulphuric acid in the fibre. Thorough rinsing, oxidation and a mild alkaline aftertreatment reduce this risk; it is a standard viva and examination point for this class.

5. Procedure C — azoic colour on cotton (naphtholation, diazotisation, coupling)

System: coupling component — a Naphthol AS type, 1.5% owf; diazo component — a Fast Base (for example Fast Red B Base), used in calculated excess over the naphthol; LR 1:20 for impregnation. Work cold throughout: diazonium salts decompose rapidly when warm.

1. **Naphthol solution:** paste 0.15 g Naphthol AS with a small volume of hot water and NaOH solution (caustic soda about 1.2–1.5 times the naphthol weight, as directed for the product) until a clear sodium naphtholate solution forms; a little Turkey red oil or wetting agent assists pasting. Make up to 200 mL, keeping the solution slightly alkaline and clear.
2. **Naphtholation:** enter 10 g pre-wetted cotton and impregnate for 20–30 min at room temperature with gentle movement. The naphtholate has substantivity for cellulose and distributes through the fibre. Lift, drain and lightly squeeze to remove excess surface naphthol without washing it out — surface naphthol later gives loose surface colour and poor rubbing fastness.
3. **Diazotisation (prepared in parallel, in an ice bath):** dissolve the Fast Base in dilute hydrochloric acid, cool to 0–5 °C with ice, and add sodium nitrite solution slowly, keeping the temperature below 5 °C and a slight excess of nitrous acid (starch–iodide paper just turns blue). Then neutralise the excess mineral acid with sodium acetate to pH about 4–5 for coupling.

$$Ar–NH_2 + NaNO_2 + 2\ HCl \rightarrow Ar–N_2^+\ Cl^- + NaCl + 2\ H_2O\ (0–5\ °C)\ \cdot\ Ar–N_2^+ + \text{naphtholate} \rightarrow Ar–N{=}N–Ar'\ \text{(insoluble azo pigment, in fibre)}$$

4. **Coupling / development:** enter the naphtholated fabric into the cold diazo bath. Colour develops almost immediately as the diazonium ion couples at the activated position of the naphthol inside the fibre. Work 15–20 min, keeping the bath cool.
5. Remove, rinse cold, then soap at the boil (2 g/L detergent, 10 min) to remove surface pigment, rinse and dry. The resulting red is brilliant with excellent wash fastness because the pigment is physically trapped inside the fibre; rubbing fastness is only moderate, which is the known weakness of the class.

6. Procedure D — disperse dye on polyester (high-temperature method)

Recipe: disperse dye 1.5% owf; dispersing agent 1 g/L; acetic acid to pH 4.5–5.0; LR 1:20; 130 °C, 45 min in a sealed HT beaker.

1. Weigh 5 g scoured polyester. Disperse 0.075 g dye: paste with dispersing agent solution and warm water, then dilute — never add dry disperse powder directly to the bath.
2. Make up to 100 mL (LR 1:20), add dispersing agent 1 g/L (0.10 g total) and set pH 4.5–5.0 with acetic acid. An alkaline bath hydrolyses many disperse dyes and shifts shade; a too-acid bath can destabilise the dispersion.
3. Enter the fabric at 50 °C, seal the beaker correctly and place it in the HT machine. Raise at 2 °C/min to 130 °C and hold 45 min. Below about 100 °C polyester takes up very little dye; uptake rises steeply as segmental mobility increases above the effective glass transition in water.
4. Cool to below 80 °C before opening the beaker — opening a hot pressurised beaker is both dangerous and a cause of creasing and unlevelness.
5. **Reduction clearing:** treat 15 min at 70–80 °C with caustic soda 2 g/L and sodium hydrosulphite 2 g/L to destroy loose surface dye, rinse, neutralise, rinse and dry. Clearing is what gives HT dyeings their rated wash and sublimation fastness.

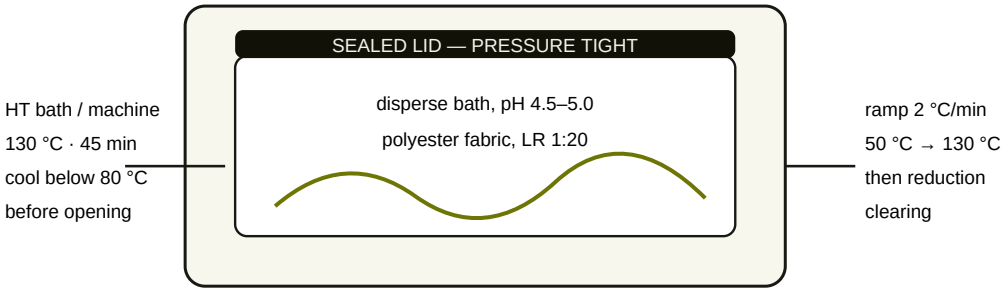


Fig. detail: sealed stainless-steel beaker — the only safe vessel for 130 °C aqueous dyeing

Fig. 2 — HT dyeing beaker schematic for the disperse dyeing of polyester (Procedure D).

7. Procedure E — pigment coloration with binder

System (padding route): pigment 3% owf equivalent in paste terms (for a 10 g sample worked by padding, prepare 100 mL pad liquor containing pigment dispersion 30 g/L, binder 100 g/L, thickener / anti-migration agent as directed, softener 10 g/L). Because pigment has no affinity, recipe is expressed per litre of pad liquor, not by exhaustion.

1. Prepare the pad liquor in order: water, binder, pigment dispersion (pre-diluted, strained), then auxiliaries. Stir gently to avoid foaming and air entrapment.
2. Pad the dry cotton at about 70% pick-up (one dip, one nip), keeping the nip pressure even; uneven pick-up prints through directly as uneven shade because nothing migrates afterwards.
3. Dry at 100–110 °C without over-drying, then **cure at 150 °C for 3 min**. During curing the binder coalesces and crosslinks into a continuous film that mechanically locks the pigment particles to the fibre surface.
4. Assess handle alongside shade: pigment coloration is fibre-independent and needs no washing, but excess binder stiffens the fabric and rubbing fastness depends entirely on film quality and curing completeness. Under-cured samples fail rubbing and washing tests immediately.

8. Worked calculation (record in full)

The reference calculation for this sheet is the reactive dyeing (Procedure A). Every other bath is costed by the same two formulas, and the examiner will ask for them.

$$\text{Dye (g)} = \text{fabric (g)} \times \%\ \text{owf} \div 100 \quad \cdot \quad \text{Chemical (g)} = \text{g/L} \times \text{bath volume (L)} \quad \cdot \quad \text{Bath (mL)} = \text{fabric (g)} \times \text{LR}$$

ITEM	CALCULATION	AMOUNT USED

Reactive dye, 2.0% owf on 10 g	10 × 2.0 ÷ 100	**0.20 g**; from 1% stock (10 mg/mL) = **20.0 mL**
Bath, LR 1:20 on 10 g	10 × 20	**200 mL**
Glauber's salt, 40 g/L	40 × 0.200	**8.0 g**
Soda ash, 15 g/L	15 × 0.200	**3.0 g**
Sulphur dye, 3% owf on 10 g (Proc. B)	10 × 3 ÷ 100	**0.30 g**; Na_2S 3% owf = **0.30 g**; soda ash 2 g/L = **0.40 g**; salt 10 g/L = **2.0 g**
Disperse dye, 1.5% owf on 5 g (Proc. D)	5 × 1.5 ÷ 100; bath 5 × 20	**0.075 g**; bath **100 mL**; dispersing agent 1 g/L = **0.10 g**
Reductive stripping bath (Sec. 11)	6 g/L and 4 g/L in 0.200 L	Hydrosulphite **1.2 g**; NaOH **0.80 g**

Check habit: always convert the bath to litres before multiplying by g/L. The commonest laboratory error is to multiply g/L by millilitres. A second check is to confirm that the % owf figure is taken on the fabric actually in that bath (5 g for the polyester, 10 g for the cotton baths in this sheet).

9. Combination shade dyeing

Most production shades are not single dyes but **combinations** — typically a trichromy (yellow + red + blue) of dyes from the *same class* with matched exhaustion and fixation behaviour. Dyes of different classes must never be mixed in one bath unless the system is designed for it: their pH, temperature and electrolyte demands conflict.

1. Select three compatible reactive dyes (same reactive group family and similar reactivity) whose combined hue matches the target.
2. Dye a first trial at the recipe proportions suggested by the supplier's shade card or by spectrophotometer prediction, using exactly the Procedure A sequence, salt and alkali profile.
3. Judge the dried, conditioned trial against the standard (Section 10). Adjust one component at a time: depth first (total dye), then hue (relative proportions).
4. Record metamerism risk: a combination that matches under D65 daylight may fail under shop (TL84) or tungsten light if its component dyes differ spectrally from the standard's dyes. Note the illuminants checked in the record.

10. Shade matching workflow

1. **Prepare comparable samples.** Trial and standard must be the same substrate, fully dried, conditioned, and presented at the same size, doubled over so no background shows through, on a neutral grey surround.
2. **View correctly.** Use a light cabinet: D65 first, then TL84 and tungsten / illuminant A to expose metamerism. View at 45° illumination / 0° viewing, without sunglasses or tinted lenses; rest the eyes on grey between judgements to avoid after-image errors.
3. **Describe the difference in words before touching the recipe:** is the trial lighter / deeper, brighter / duller, and shifted toward which hue (redder, yellower, bluer, greener)? This three-part description (depth, brightness, hue) is the professional vocabulary of the shade-matching bench.
4. **Correct systematically.** Depth error → scale all components together. Hue error → adjust the component(s) controlling that hue direction only, in small steps (about 5–10% of that component). Re-dye; do not try to correct a cotton reactive shade by topping unless the laboratory procedure explicitly allows it, since topping rarely reproduces in bulk.
5. **Confirm instrumentally where available.** A spectrophotometer reports ΔE (CIELAB) between trial and standard; a common laboratory pass criterion is $\Delta E \leq 1.0$ under D65 with no metamerism failure under the secondary illuminant. File the final recipe, conditions and ΔE with the retained sample.

11. Stripping — reductive stripping of a faulty reactive dyeing

Stripping removes or destroys colour so a faulty lot can be re-dyed. Two levels are distinguished: **shade correction (partial stripping / levelling)** and **full stripping** to a near-white ground. Reactive dyes are covalently bound, so the bond must be chemically broken: reduction with hydrosulphite in alkali is the standard laboratory method, but even good stripping rarely removes 100% of a reactive dyeing, and the fibre must be checked for strength loss before re-dyeing.

Recipe (200 mL bath, LR 1:20 on 10 g): sodium hydrosulphite 6 g/L (1.2 g), caustic soda 4 g/L (0.80 g), 80–90 °C, 20–30 min.

1. Wet out the faulty fabric thoroughly. Prepare the bath with caustic soda first, heat to 80 °C, then add hydrosulphite last (it decomposes quickly in hot aerated liquor).
2. Enter the fabric, keep it submerged and covered, and run 20–30 min at 80–90 °C. The chromophore is reduced and the dye–fibre bond is attacked under the alkaline reducing conditions; the bath takes up the destroyed colour.
3. Drain, rinse thoroughly, neutralise, and rinse again. Assess the residual shade in daylight: record the percentage removal estimated visually and whether the ground is even enough for the intended re-dye shade (pale re-dyes demand the most complete stripping).

4. Note the alternatives for the record: oxidative stripping (hypochlorite or permanganate systems) attacks the chromophore differently and is harsher on cotton; it is chosen only when reductive stripping is insufficient and fibre damage is acceptable.

12. Results and observations

DYEING	EXPECTED RESULT	OBSERVATIONS TO RECORD
Reactive / cotton	Deep, level shade; clear final soap liquor after second rinse	Exhaustion before alkali vs after fixation; colour in soap bath (hydrolysed dye); wash and rub fastness impression
Sulphur / cotton	Dull deep shade; bronziness if oxidised at surface	Bath odour, leuco colour, shade change on oxidation, any tendering note
Azoic / cotton	Brilliant red developing instantly on coupling	Development speed, surface colour removed at soaping, rub fastness limitation
Disperse / polyester	Level deep shade after clearing; pale before clearing, surface-dull	Effect of clearing on depth and brightness; levelness at seams / folds
Pigment / cotton	Shade correct immediately after curing; stiffer handle	Handle change, rub-off on white cloth (dry and wet), effect of under-curing trial
Stripping	Large shade loss, slightly uneven pale ground	Residual depth, evenness, suitability for re-dyeing, any strength change by hand tear comparison

Attach all dried samples, labelled with class, recipe reference and date, in the practical record in the order A–E, followed by the combination shade, the matched shade (trial + standard together) and the stripped sample (before + after pieces side by side).

13. Precautions

- Caustic soda, soda ash solutions, hydrochloric acid and sodium nitrite: gloves and eye protection; diazotisation is carried out cold and the diazo solution is never stored, heated or allowed to dry — dry diazonium salts can be explosive.
- Hydrosulphite: add to liquor, never the reverse onto a dry heap; weigh last, use at once, keep stripping and vatting baths covered; the powder decomposes in moist air with heat evolution.
- HT beakers: check the seal and gasket, never overfill, never open above 80 °C, and cool under the machine programme — a beaker opened hot discharges boiling acidic liquor.

- Sulphur baths release sulphurous fumes: work with ventilation and keep the face away when lifting the cover.
- Weigh dyes on a covered balance and clean spills at once; reactive and disperse dye dust contaminates every neighbouring bath and sample.

14. Practice (viva preparation)

1. Why is soda ash added only after salt exhaustion in reactive dyeing, and what competes with fixation once alkali is present?
2. Calculate the dye, salt and alkali for a reactive dyeing of 8 g cotton at LR 1:25 with dye 1.5% owf, salt 30 g/L and soda ash 12 g/L.
3. Explain, with the chemistry, why an azoic colour has high wash fastness but only moderate rubbing fastness.
4. Why can disperse dyeing of polyester not be done satisfactorily at 95 °C in an open bath, and what two aftertreatment purposes does reduction clearing serve?
5. A trial shade is deeper than standard and too red. State the two recipe corrections, in order.
6. Why does reductive stripping seldom return reactive-dyed cotton to pure white, and what must be checked before re-dyeing?

Answer key

1. Alkali both enables fixation (Cell–O^- formation) and hydrolyses dye with water. Exhausting first under neutral salt conditions builds a level, high dye concentration in the fibre; fixation then locks that distribution in. Hydrolysis competes throughout the alkaline stage, which is why time, temperature and alkali dose are controlled and hydrolysed dye is soaped off afterwards.
2. Bath = 8 × 25 = 200 mL. Dye = 8 × 1.5 ÷ 100 = 0.12 g. Salt = 30 × 0.200 = 6.0 g. Soda ash = 12 × 0.200 = 2.4 g.
3. The naphthol and diazonium components couple inside the fibre to a large, water-insoluble pigment that cannot dissolve out in washing — hence high wash fastness. Pigment deposited at or near the surface from excess naphthol is only mechanically held and is removed by abrasion — hence moderate rubbing fastness, improved by thorough boil-soaping.
4. At 95 °C polyester chain mobility is too low and dye solubility / diffusion too slow for practical exhaustion in reasonable time; 130 °C opens the structure sufficiently. Reduction clearing (a) destroys loose surface dye, restoring brightness, and (b) thereby delivers wash, rub and sublimation fastness.
5. First reduce total dye to correct depth (scale all components down proportionally). Then, once depth matches, reduce the red component (or increase its complement) in a small step to correct hue.

Depth is always corrected before hue, because hue judgement is unreliable at the wrong depth.

6. The dye is covalently bonded; reduction must both destroy the chromophore and cleave dye–fibre bonds, and some bound fragments and residual colour persist, especially in deep shades. Check residual evenness against the intended re-dye depth and check fibre strength, because hot alkaline reduction can weaken cotton.

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